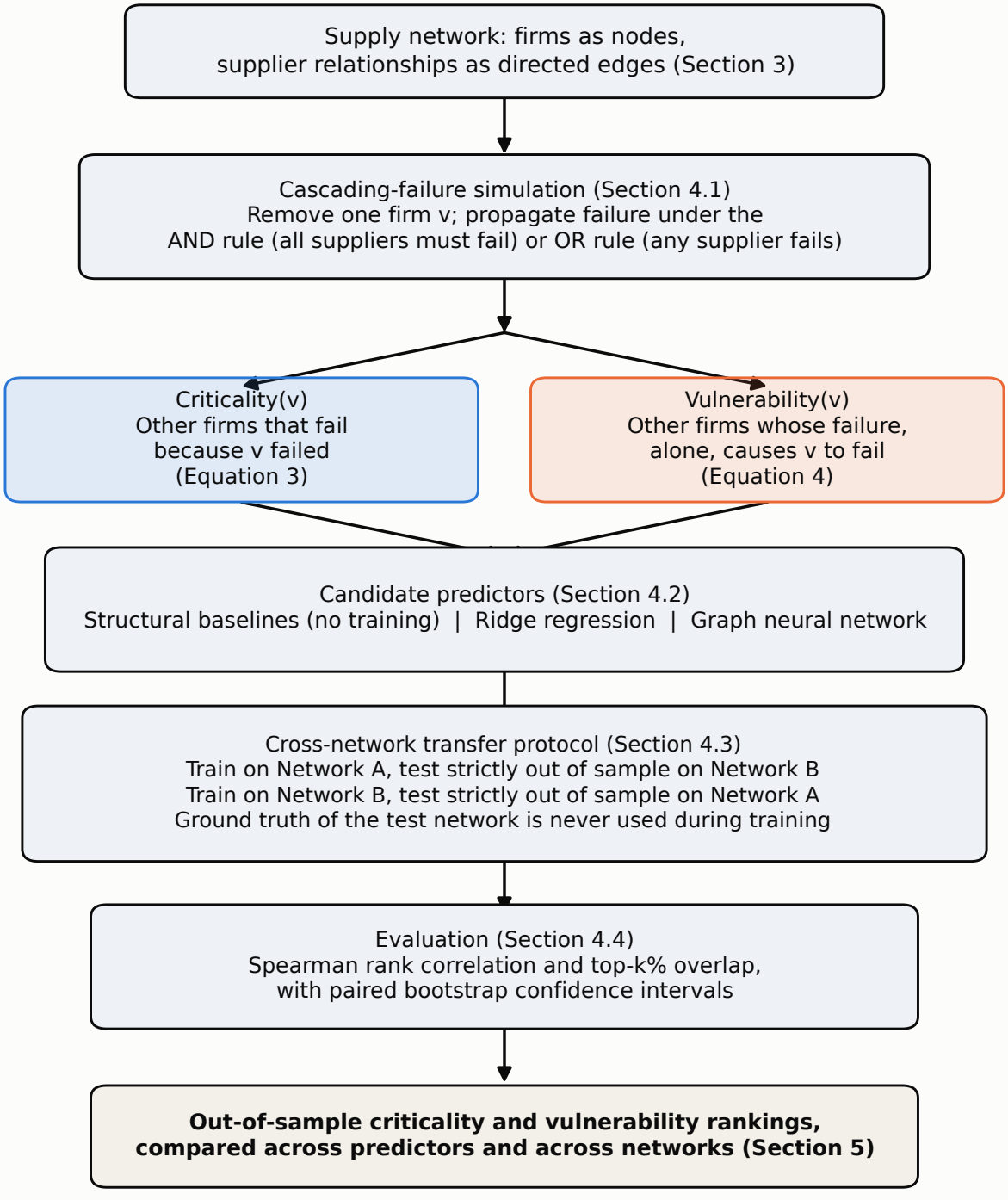
Supply network: firms as nodes,
supplier relationships as directed edges (Section 3)
Cascading-failure simulation (Section 4.1)
Remove one firm v; propagate failure under the
AND rule (all suppliers must fail) or OR rule (any supplier fails)
Criticality(v)
Other firms that fail
because v failed
(Equation 3)
Vulnerability(v)
Other firms whose failure,
alone, causes v to fail
(Equation 4)
Candidate predictors (Section 4.2)
Structural baselines (no training) | Ridge regression | Graph neural network
Cross-network transfer protocol (Section 4.3)
Train on Network A, test strictly out of sample on Network B
Train on Network B, test strictly out of sample on Network A
Ground truth of the test network is never used during training
Evaluation (Section 4.4)
Spearman rank correlation and top-k% overlap,
with paired bootstrap confidence intervals
Out-of-sample criticality and vulnerability rankings,
compared across predictors and across networks (Section 5)